



Docket No. 2015-6

Study on Software-Enabled Consumer Products

**Initial Comments of the Software and Information
Industry Association**

February 16, 2016

The Software & Information Industry Association (“SIIA”) respectfully submits these comments to the United States Copyright Office in response to the Notice of Inquiry entitled “Software-Enabled Consumer Products Study,” 80 Fed. Reg. 77668 (December 15, 2015).

About SIIA

SIIA is the principal trade association of the software and information industries and represents over 800 companies that develop and market software and digital content for business, education, consumers, the Internet, and entertainment. SIIA’s members range from start-up firms to some of the largest and most recognizable corporations in the world, and one of SIIA’s primary missions is to protect their intellectual property and advocate a legal and regulatory environment that benefits the software and digital content industries. SIIA member companies are market leaders in many areas, including but by no means limited to:

- software publishing, graphics, and photo editing tools
- corporate database and data processing software
- financial trading and investing services, news, and commodities exchanges
- online legal information and legal research tools
- protection against software viruses and other malware and
- education software, online education services and open source software.

Introduction

As Moore's Law has made computers smaller and smaller, our members have seen increasingly vibrant and powerful versions of their publications and software delivered on a variety of consumer devices. Cell phones, for example, contain feature-rich versions of photo editing, database, and similar software that only existed on desktops just a few years ago. Devices that look like speakers are capable of accessing the internet, responding to spoken queries, and returning information.

The devolution of computing power into smaller and smaller devices and the promotional buzz surrounding the so-called "Internet of Things" (IoT) have raised curiosity among policy makers about whether to change existing law with respect to what some have unfortunately termed "embedded software," and which the Office has termed "copyrighted software in everyday products." The Department of Commerce's Internet Policy Task Force recently stated that "the case has not been made in the record to change the law to address the sale of [software contained in] products or devices,"¹ a similar conclusion to that of the Copyright Office fifteen years ago.²

SIIA shares that conclusion. SIIA believes that no change to existing law is necessary to address embedded software, software contained in consumer goods, the "Internet of Things," or whatever moniker is ultimately used to describe it. That is because we respectfully believe that this particular study is proceeding from a false dichotomy: there is no such distinction between "embedded" and other kinds of software. There is merely

¹ Department of Commerce Internet Policy Task Force, [White Paper on Remixes, First Sale and Statutory Damages](#) at 65 (Jan. 2016) ("IPTF White Paper").

² *E.g.*, U.S. Copyright Office, DMCA Section 104 Report (2001).

software—some of which is licensed, and some of which is sold.

Most SIIA members license their products.³ The law permits (and should continue to permit) technology providers to choose the models that best fit their markets. The innovative landscape of electronic commerce and high-tech products derives from the wide availability of software and content, and flexibility of legal doctrines that allows providers to tailor their offerings to customer demand. Licensing permits a software publisher to offer a fully functional “academic” version of its product to students at a deeply reduced price, but the rights granted do not permit use for commercial purposes.⁴ Similarly, “OEM licenses” bundle software with, or install software directly on scanners or desktop computers, and require the software to be used and distributed only with that hardware. Often, the hardware manufacturer was granted a deep discount as part of the OEM license terms. Another example is “site licenses,” which are defined by some geographic restriction on use, such as a specific company, area, or even department or floors of a building.⁵

Licenses have enabled SIIA members to tailor products for specific customer needs, adjusting features, benefits,

³ Licensing is the dominant method of software distribution. See Software & Information Industry Association, *Software and Information: Driving the Knowledge Economy* (January 24, 2008) at 7-8, <http://www.sii.net/estore/globecon-08.pdf>.

⁴ See, e.g., *ProCD, Inc., v. Zeidenberg*, 86 F.3d 1447 (7th Cir. 1996) (“Instead of tinkering with the product . . . [software companies] turned to the institution of contract.”); id. at 1455 (“Terms and conditions offered by contract reflect private ordering, essential to the efficient functioning of markets.”).

⁵ See, e.g., Software & Information Industry Association & LicenseLogic LLC, *Certified Software Manager Student Manual* (SIIA Publications 2004) at 4:1–4:52.

rights, and price according to the needs of each customer base rather than a “one size fits all” model – a model which logically could require a higher price.⁶ Consequently, licensing provides benefits to consumers that section 109 cannot provide.

Those advantages are not limited to software. For example, the licensing of online and digital versions of textbooks provides numerous benefits to students and teachers. They often come with special features such as embedded quizzes, electronic flash cards, the ability to share notes online with fellow students and/or embedded links to videos and articles from a professor’s lectures. These digital texts may also allow the teacher to monitor a student’s progress, the amount of time the student spends reviewing the material, and the student’s performance on the embedded quizzes. The teacher can then use this information to determine what material the student finds difficult and tailor study plans to keep the student on the right track. The nature of digital textbooks renders updating and editing them a much faster and easier task than their analog counterparts, and revisions can be distributed to users almost immediately—all without the additional expense of printing, handling, and processing returns.

In addition, institutions increasingly *want* course materials delivered “inside” their digital classrooms so they can ensure that all students have access to the same materials and they can see how each student is performing.

⁶ Cf. Robert Maier, [Penguin Random House Now Licensed for Perpetual Access](#) (Jan 2016) (“Though we were in shock when HarperCollins instituted its 26-loan limit in 2011, that rental model is now recognized as an attractive alternative to high-priced perpetual access. At this point, the ideal arrangement would be the ability to license any title for perpetual access and to license the same title for a set number of circulations or a set time period, combining the library’s need to build its collection and to meet current popular demand.”).

These innovative products and business models would not be possible if textbook publishers were no longer able to rely on licensing.⁷

SIIA's comments in this study are greatly informed by our members' marketplace experience. Software and information providers have also begun to provide access to technology and information services in which the consumer can acquire access to far more content and functionality than she could have in tangible goods. And that access comes as cloud computing has already changed consumer relationships with both software and information from a good to a service. These services increasingly seek to provide a seamless user experience across on PCs, tablets, smartphones, televisions, automobiles, household appliances, and other devices. Instead of purchasing a copy of photo-editing software that is downloaded onto a personal computer, a consumer sends copies of photographs to the cloud to be edited and stored, and then returned for saving, printing, or display. Similarly, a variety of software is finding its way into seemingly "ordinary" devices, such as pens that can capture notes as written, and then send those notes to the cloud for storage, character recognition, and export into SIIA member software.⁸ In these (and

⁷ The result of all these evolving business models has been a dramatic drop in student textbook spending from \$192 in the fall of 2008 to \$138 in the spring of 2013. See Stephanie Simon and Madeline Will, *Textbook publishers revamp ebooks to fight used market*, 4-traders.com (July 23, 2013) at <http://www.4-traders.com/PEARSON-PLC-4000637/news/Textbook-publishers-revamp-ebooks-to-fight-used-market-17119420/>. (This is despite claims that the cost of new textbooks rises about 6 percent a year. See GAO Report 13-368, *College Textbooks: Students Have Greater Access to Textbook Information* at 6 (June 2013) at <http://www.gao.gov/assets/660/655066.pdf>.

⁸ E.g., <http://digital-pen-review.toptenreviews.com/> [find others]. [See also <http://www.livescribe.com/en-us/legal/#TermsSales>].

many other) situations, the relationship between consumer and software is not the relationship between a consumer and an inert copy; it is the relationship between a consumer and a live system.

SIIA opposes any change to existing law to deal with so-called “embedded software.” The flexibility of existing law both enables our members to make their software and information products more widely available and permits them to maintain the integrity of their products and services. This principle of product integrity protects not only the public interest in copyright’s economic incentive and healthy marketplace competition, but also broader health and safety interests. Maintenance of product integrity is essential to the full development of the IoT’s economic and social potential as it ensures that devices will operate as the consumer expects, and device manufacturers intend. In many cases, ensuring product integrity will require users to abide by the terms of software licenses and other contractual obligations.

Our more detailed responses to the NOI’s questions follow below, but can be summarized in the following basic themes:

- **The Copyright Act has successfully created a baseline of property rights, and the state law of contract fills in the gaps.** Against that backdrop, software development has flourished, and the industry is rapidly evolving in response to technological change. The resulting market ecosystem has spawned frenetically paced innovation and development while maintaining the incentives to create that the copyright law provides.
- **The category of software described in the NOI is one that does not exist and cannot be constructively defined.** Capital is flowing into the creation of “smart goods,” products and devices made more effective or “intelligent” through the incorporation of software and often

an Internet connection. There is no evidence that innovation is being harmed by this activity or the licensing terms that accompany this software. Proposals to create legislative definitions of “embedded software,” “software contained in consumer goods” is doomed to failure, and will have negative, if unintended, consequences.

- **To the extent that concerns have arisen over the development of secondary markets, such concerns are beyond the scope of this rulemaking.** Issues relating to so-called “rights to tinker,” the “right to repair” and other related issues are beyond the scope of this rulemaking as they assume ownership of a copy. Such concerns (to the extent they legitimately exist) are best addressed within the context of the Office’s examination of section 1201.

RESPONSES TO THE NOI

I. Existing Law Facilitates the Proliferation of Numerous Business Models (Question 1, 2)

Under existing law, software is a “computer program;” namely, “a set of statements or instructions to be used directly or indirectly in a computer in order to bring about a certain result,” and is treated as a subset of literary works.⁹ It provides the owners of copyright in computer programs with the same rights as any other copyright owner, but subject to a few specific limitations. First, section 117 permits the owner of a copy to adapt a particular work and to make additional copies as an “essential step” in the operation of a machine, or for archival purposes.¹⁰ It also permits the owner or lessee of a machine to copy software without infringing the

⁹ 17 U.S.C. § 101.

¹⁰ *Id.* § 117 (a) (1), (2).

copyright for purposes of maintenance or repair.¹¹ In addition, Congress crafted an exemption from software's general rental right in section 109(b)(1)(B)(i) that allows rental, lease, or lending of devices that cannot be copied during the ordinary operation of a machine.¹²

Each of these provisions contemplates the licensing of copyrighted works. Section 117(a) extends certain archival and modification rights, but only to the owner of a copy. Section 117(c) limits the reproduction right, but does not invalidate license terms.

Similarly, section 109 makes plain that its limitation on the distribution right does not apply to a copy of the work through "rental, lease or lending."¹³ And, although section 109 allows the owner of a machine to lease the machine without violating the distribution right, that limitation has no relevance when the reproduction right is expressly licensed. The text and structure of the act create a baseline of freely alienable property rights that can be rented, licensed, or transferred in gross in a variety of ways.

¹¹ *Id.* § (c), (d); *see also* H.R. Rep. No. 105-551, at 26-27 (1998) (stating a "narrow and specific intent of relieving independent service providers, persons unaffiliated with either the owner or lessee of the machine, from liability under the Copyright Act when, solely by virtue of activating the machine in which a computer program resides, they inadvertently cause an unauthorized copy of that program to be made.").

¹² *See also* TRIPS, art. 11 (exempting signatories from providing rental right for computer software when the software is not the essential purpose of the rental). Like any copyrighted work, software is subject to the idea/expression dichotomy, an area that SIIA believes the courts have well in hand.

¹³ 17 U.S.C. § 109(d); *id.* § 117 (a)(1), (2). *See also id.* § 202 (transfer of physical object does not convey intellectual property rights, and vice versa); H.R. Rep. No. 94-1476, at 80, 124-25 (1976) (describing limits of sections 109 and 202).

A. The Market for Software in Smart Goods is Healthy and Growing

Against this backdrop, the market for software and information generates billions in revenues and continues to grow.¹⁴ The framework under which that growth has occurred operates from a fairly simple assumption: copyright provides property rights, which the owner may license (or not), or sell (or not), in response to market demand. A transaction for software could involve Article 2 of the Uniform Commercial Code, the common law of service contracts, as well as intellectual property doctrines including copyright and patent law.

That model has extended quite well to so-called “smart goods.” Between 2010 and 2014, funding for IoT startups has more than doubled, and attracted about \$7.5 billion in investment,¹⁵ and the number of connected devices is estimated to grow by thirty percent in 2016.¹⁶ That growth comes as many of SIIA’s member software companies’ relationship has changed from change from an installed user base in which customers license software CDs and install that software on computers to one in which the software is hosted and run remotely, and sold as a service. The shift reflects customer demand for both software and information that is automatically updated and accessible in a convenient form.

¹⁴ See Software & Information Industry Association, *The U.S. Software Industry: An Engine for Economic Growth and Employment* (2014) <http://www.siiia.net/Admin/FileManagement.aspx/LinkClick.aspx?fileticket=yLPW0SrBfk4%3d&portalid=0>. See also generally *Siwek, The Copyright Industries and the U.S. Economy* (2014).

¹⁵ CB Insights, [Funding to IoT Startups Has More Than Doubled in Six Years](#) (November 10, 2016).

¹⁶ Gartner, Press Release, [Gartner Says 6.4 Billion Connected “Things” Will Be in Use in 2016, Up 30 Percent From 2015](#) (Nov. 10, 2015).

That same kind of demand for convenience has entered into consumer goods, and has begun to change the way that individuals interact with devices. Such devices collect information about their users and tailor functionality to a user's experience. Cars, networked thermostats, activity monitors, and even refrigerators have begun incorporating "smart" features—some of which will be enormously successful, and some of which will not.

II. Flexible Licensing Terms Ensure Health, Safety, and Product Integrity (Questions 1,2,3,5)

The licensing of software contributes to product integrity: the operation of the product in the manner in which the consumer expects, and the manufacturer intends. License terms may restrict the conditions under which the user may modify the software and the manner in which it interacts with a network. The manufacturer and the software provider should be permitted to decide whether or not the software in a consumer good has been licensed, sold, or is the subject of a service contract. Depending on the facts of the transaction (and the vagaries of state law), software can be a sale, a license, or a contract for services, or a combination of all three.¹⁷ The distinction

¹⁷ See, e.g., *Simulados Software, Ltd. v. Photon Infotech Private, Ltd.*, 40 F. Supp. 3d 1191, 1199-200 (N.D. Cal. 2014); (noting that courts look to the essence of the agreement to determine whether the UCC applies); *Vernor v. Autodesk, Inc.*, 621 F.3d 1102, 1111 (9th Cir. 2010) (establishing three-part test to determine whether a transaction for software is a sale or a license); *DSC Commc'ns Corp. v. Pulse Commc'ns, Inc.*, 170 F.3d 1354, 1362 (Fed. Cir. 1999) (applying a multi-factor test to determine whether a licensee is in fact the "owner" of a copy); *MDY Indus., LLC v. Blizzard Entm't, Inc.*, 629 F.3d 928, 939 (9th Cir. 2010), *as amended on denial of reh'g* (Feb. 17, 2011), *opinion amended and superseded on denial of reh'g*, No. 09-15932, 2011 WL 538748 (9th Cir. Feb. 17, 2011) (noting difference between covenants and conditions).

that the NOI purports to draw between “everyday consumer products” and “non-everyday” products would destroy both software providers and manufacturers’ abilities to protect product integrity and is one that the copyright law does not, cannot, and should not draw.

A. Disruption of Licensing Law Will Create Negative Business, Health, and Safety Consequences.

For example, companies are creating ingestible medical sensors that connect with a patch worn on the skin. The sensor communicates with the patch to send blood pressure, medication, and other information to the cloud, and can help medical professionals determine how well a particular medication regime is working.¹⁸ The collected data is contained by software on the ingestible “pill,” more software on the patch, a software application for the patient, and in a portal for health care professionals. We respectfully suggest that a copyright rule mandating that a particular product is automatically “sold” is unwise public policy.

SIIA members also use and engage in the use of open-source software. Although these licenses permit free transfer and reproduction of the software, licensees must disclose improvements and modifications that they make.¹⁹ These obligations do not run to the “owner” of a copy—only to licensees. Not only is open source software frequently included in consumer goods, but its use is a large part of IoT development.²⁰ Uncertainty over the enforceability of such licenses due to the transfer of the goods that happen to contain licensed software would be highly disruptive to

¹⁸ See, e.g., <http://www.proteus.com/how-it-works/> (visited 3 February 2016).

¹⁹ E.g., <https://opensource.org/licenses/GPL-3.0> §§ 5, 6, 7.

²⁰ Giles Gravier, [Open Source as a Driver of Internet of Things](#), InfoQ (Jan. 9, 2016); Steven Max Peterson, [Google Readies Developers to Build the IoT, Starting with its Open Version of Bluetooth Low Energy Beacons](#), Network World, Jan 12, 2016).

those entities that have elected to pursue an open source model of software development.

No matter which licensing model is chosen—open, closed, a combination of both, or none—existing law has created a wealth of legitimate business models and technologies. To the extent that *owners* of copies of computer software are prevented from repairing particular devices by technological protection measures within the scope of section 1201, that discussion is beyond the scope of this NOI. SIIA remains puzzled regarding the specific problem that legislation addressing embedded software would allegedly solve.

B. Existing Law Has Created Healthy Secondary Markets

That absence of evidence, however, has not prevented claims that the spread of embedded software will result in the deterioration of secondary markets. Such claims proliferated when the *ProCD* decision was decided nearly twenty years ago. Due to the presence of software and other digital information in hard goods, the argument went, all kinds of secondary transactions in hard goods would be prohibited. Experience suggests treating those claims with considerable skepticism.

The “Craigslist apocalypse” simply has not happened, for several reasons. First, prophecies of doom notwithstanding, a simple look around eBay and many other platforms reveals that robust secondary markets exist for all kinds of goods, and are likely to remain so. Consumers have certain expectations regarding both transferability and interoperability, and the marketplace has generally provided sufficient incentives to meet those expectations.²¹ An expensive purchase such as a car, or even smart watch, is worth far less if it cannot be resold.

²¹ *Cf., e.g.,*

<http://www.developers.meethue.com/content/friends-hue-program-update> (noting how Philips changed firmware with its

Second, although the law permits software to be licensed and not sold, it hardly requires it. The courts require licensors to take certain steps to make clear to subsequent purchasers that software is licensed, and there must be sufficient indicia of licensing to support that conclusion.²² In the event that “everyday consumer products” were subject to license terms that restricted their transfer in secondary markets, courts would likely scrutinize those transactions using the indicia of sale and licensing developed in the common law and interpretive decisions of the copyright act, as appropriate. It is far more likely that the situation will continue as it is now: goods will be sold containing some software that is owned, and some that is licensed.²³ The scope of those licenses (and sales) will vary by industry.

However, it is possible that the rapid pace of change may outrun existing consumer expectations. Although we have antitrust concerns about how its suggestions might be implemented in practice, we do believe that the IPTF’s recommendation of a multi-stakeholder process to discuss a consumer education campaign and possible best practices could provide a useful way forward.²⁴

hubs to ensure interoperability after consumer outcry); Treehouse Foods, Inc. Reports Second Quarter Results, <http://phx.corporate-ir.net/phoenix.zhtml?c=191105&p=irol-newsArticle&ID=1956509> (Aug 7, 2014) (Green Mountain coffee competitor announcing successful reverse engineering around Keurig blocking technology).

²² *E.g.*, *UMG Recordings, Inc. v. Augusto*, 628 F.3d 1175, 1180 (9th Cir. 2011) (finding that sticker on CD insufficient to create a license); *Adobe Sys. Inc. v. Christenson*, 809 F.3d 1071, 1080 (9th Cir. 2015) (first sale defense upheld because copyright owner did not bring forth sufficient proof of licensing).

²³ *See, e.g.*, <http://www.belkin.com/us/end-user-license-agreement/> (permitting transfer of software in smart goods provided originals deleted).

²⁴ IPTF White Paper, *supra* note 1, at 69.

III. Conclusion

In short, while we believe this kind of review is entirely wise and appropriate and we applaud the Office's diligence, the law needs no changes to address issues in software copyright.

Respectfully submitted,

/s/

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